

SciRet: A Compute-Aware Empirical Study of Retrieval and Reranking for Scientific RAG



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Keywords: retrieval-augmented generation, scientific question answering, hybrid retrieval, BM25, BGE-M3, reciprocal rank fusion, cross-encoder reranking, CORD-19, domain mismatch, RAGAS

Abstract:

We introduce SciRet, a compute-aware empirical study of retrieval-augmented generation for scientific question answering over CORD-19. Rather than proposing a new model, we evaluate a fixed scientific RAG pipeline across three corpus scales: 1,034 chunks (1K papers), 5,160 chunks (5K papers), and 15,480 chunks (15K papers). The pipeline combines sentence-window chunking, BM25, BGE-M3 dense retrieval, reciprocal rank fusion, optional cross-encoder reranking, and grounded answer generation. Across these settings, hybrid retrieval is more robust than either sparse-only or dense-only retrieval in our setting, reaching Recall@10 of 1.000 at 1K and 15K. In contrast, an MS MARCO-trained cross-encoder reranker reduces precision on the scientific corpus, suggesting that domain mismatch can outweigh the benefits of stronger query-passage interaction. Generation faithfulness measured with RAGAS increases with corpus scale in our setup. Retrieval evaluation uses pseudo-relevance labels derived from the hybrid system, so we treat the results as controlled comparative evidence rather than a benchmark claim. We release code, indexes, and evaluation outputs to support replication and follow-up studies.

Paper Type: Short

Research Area: Information Retrieval and Text Mining

Research Area Keywords: Information Retrieval and Text Mining, Machine Learning for NLP, Question AnsweringInformation Retrieval and Text Mining, Question Answering, Resources and Evaluation

Contribution Types: NLP engineering experiment, Approaches to low-compute settings (efficiency), Data analysis

Languages Studied: English

EMNLP 2026 AI Reviewing Experiment: no

TLDR: Hybrid BM25+BGE-M3 retrieval outperforms sparse and dense methods alone for scientific RAG over CORD-19, while an MS MARCO cross-encoder reranker consistently hurts precision due to domain mismatch.

Reassignment Request Area Chair: This is not a resubmission

Reassignment Request Reviewers: This is not a resubmission

Preprint: yes

Preprint Status: There is no non-anonymous preprint and we do not intend to release one. (this option is binding)

Preferred Venue: EMNLP

Visa Needs: yes

Country Of Origin: BD

Consent To Share Data: yes

Consent To Share Submission Details: On behalf of all authors, we agree to the terms above to share our submission details.

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: Yes

A2 Elaboration: Section 6 (Limitations) and the Ethics Statement discuss potential risks, including the risk of generated answers being used for medical decision-making without expert review, and the limitations of pseudo-relevance labels and small query sets.

B Use Or Create Scientific Artifacts: No

B1 Cite Creators Of Artifacts: Yes

B1 Elaboration: Section 2 (Related Work) and the References cite all artifact creators, including CORD-19 (Wang et al., 2020), BGE-M3 (Chen et al., 2024), BM25 (Robertson and Zaragoza, 2009), and RAGAS (Es et al., 2023).

B2 Discuss The License For Artifacts: No

B2 Elaboration: Licenses for used artifacts (CORD-19, BGE-M3, BM25Okapi, RAGAS) were not explicitly discussed in the paper. All artifacts used are publicly available under open licenses (Apache 2.0, MIT, or CC BY) and were used strictly for research purposes within their intended scope.

B3 Artifact Use Consistent With Intended Use: Yes

B3 Elaboration: Section 3 (Method) discusses the use of CORD-19 for research purposes only, consistent with its intended academic use. The cross-encoder reranker (MS MARCO-trained) is used for evaluation rather than deployment, consistent with its research-oriented release. Released code and indexes are intended for replication and follow-up research only.

B4 Data Contains Personally Identifying Info Or Offensive Content: N/A

B4 Elaboration: CORD-19 is a dataset of scientific papers — no personally identifying information or offensive content involved.

B5 Documentation Of Artifacts: Yes

B5 Elaboration: Section 3.1 (Corpus and Scale Protocol) and Table 1 document the CORD-19 dataset used, including corpus scales (1K, 5K, 15K papers), chunk counts, mean chunk tokens, and evaluation query counts. The Appendix provides additional chunking analysis and compute budget details.

B6 Statistics For Data: Yes

B6 Elaboration: Table 1 (Section 3.1) reports dataset statistics including papers indexed, chunk counts, mean chunk tokens, and evaluation queries across all three scales (1K, 5K, 15K). The Appendix (Table 5) provides additional chunking strategy statistics.

C Computational Experiments: Yes

C1 Model Size And Budget: Yes

C1 Elaboration: Appendix G (Compute Budget) reports wall-clock time for each pipeline stage (chunking, embedding, BM25 indexing, retrieval, reranking, generation) across all three scales. The paper uses BGE-M3 (570M parameters) for dense retrieval and cross-encoder/ms-marco-MiniLM-L-6-v2 for reranking, run on available GPU hardware.

C2 Experimental Setup And Hyperparameters: Yes

C2 Elaboration: Section 3.2 (Retrieval and Reranking) describes the experimental setup including RRF parameter (k=60), Stage 1 candidate count (50 per retriever), and retrieval cutoffs (K=1,3,5,10,20). No hyperparameter search was performed; all settings were fixed across scales by design to isolate scale effects.

C3 Descriptive Statistics: No

C3 Elaboration: Results are reported from single runs across each scale due to the compute-aware nature of the study and the fixed pipeline design. No error bars or repeated runs are reported. This is acknowledged as a limitation in Section 6, where we note that the 15-query evaluation set limits statistical power.

C4 Parameters For Packages: Yes

C4 Elaboration: Section 3.2 and 3.3 report key package settings: BM25Okapi for sparse retrieval, BGE-M3 for dense embeddings, RRF with k=60, cross-encoder/ms-marco-MiniLM-L-6-v2 for reranking, and RAGAS 0.2.6 with GPT-4o-mini as the judge model for generation evaluation.

D Human Subjects Including Annotators: No

D1 Instructions Given To Participants: N/A

D2 Recruitment And Payment: N/A

D3 Data Consent: N/A

D4 Ethics Review Board Approval: N/A

E AI Assistants In Research Or Writing: Yes

E1 Information About Use Of AI Assistants: Yes

E1 Elaboration: The Ethics Statement includes a Generative AI Disclosure paragraph stating that Claude and ChatGPT were used to assist in refining prose, improving readability, and copyediting select portions of the manuscript. The core ideas, technical methodology, and experimental evaluations were fully developed by the authors.

Author Submission Checklist: yes

Association For Computational Linguistics - Blind Submission License Agreement: On behalf of all authors, I agree

Submission Number: 8722

Discussion

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Official Review of Submission8722 by Reviewer 5KS8

Official Review by Reviewer 5KS8 05 Jul 2026, 09:41 (modified: 08 Jul 2026, 18:56) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 5KS8, Authors

Revisions

Paper Summary:

This paper presents SciRet, a compute-aware empirical study of retrieval and reranking components within a fixed scientific RAG pipeline over subsets of the CORD-19 corpus at three scales (1K, 5K, 15K papers).

Summary Of Strengths:

- The paper frames a practical, compute-aware comparison across retrieval and reranking choices for scientific RAG, emphasizing controlled design and reproducibility rather than proposing a new model.
- The paper is generally clear, with a concise description of the pipeline, explicit hyperparameters for RRF, and tabulated results.

Summary Of Weaknesses:

- Pseudo-relevance labels are derived from the hybrid retriever being evaluated, creating circularity that undermines the validity of both recall and precision comparisons—especially the “negative reranking” conclusion.
- No evaluation against established scientific IR/QA benchmarks with human relevance labels (e.g., TREC-COVID, SciFact, BioASQ), nor use of pooled labeling, which would mitigate circularity.
- Insufficient detail on the 15-query set origin, exact overlap across scales, and prompt/generation settings (e.g., number of retrieved passages, truncation, and citation mapping details).

Comments Suggestions And Typos:

- How exactly were pseudo-relevance labels constructed at each scale, and are they system-specific per scale or fixed across scales? Can you reconcile why hybrid R@3 is not 1.0 at 1K/5K if the top-3 hybrid results define relevance?
- Are the 15 queries identical across the 1K/5K/15K scales? How were they sourced and validated, and do they have any topical skew that could favor sparse or dense retrieval?

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 4 = Strong: This study provides sufficient support for all of its claims. Some extra experiments could be nice, but not essential.

Excitement: 4 = Exciting: I would mention this paper to others and/or make an effort to attend its presentation in a conference.

Overall Assessment: 2.5 = Borderline Findings

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 4 = Useful: I would recommend the new datasets to other researchers or developers for their ongoing work.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.



Official Review of Submission8722 by Reviewer SPUZ

Official Review by Reviewer SPUZ 02 Jul 2026, 18:24 (modified: 08 Jul 2026, 18:56) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer SPUZ, Authors

Revisions

Paper Summary:

This paper presents SciRet, a compute-aware empirical study of retrieval and reranking for scientific question answering using the CORD-19 dataset. Rather than proposing a new model, the authors evaluate a fixed RAG pipeline across three corpus scales: 1K papers (1,034 chunks), 5K papers (5,160 chunks), and 15K papers (15,480 chunks). The pipeline combines sentence-window chunking, BM25 sparse retrieval, BGE-M3 dense retrieval, reciprocal rank fusion (RRF), optional cross-encoder reranking (MS MARCO-trained), and GPT-4o-mini-based grounded answer generation. The key findings are: (1) hybrid retrieval (BM25 + BGE-M3) is more robust than either sparse-only or dense-only retrieval, reaching Recall@10 of 1.000 at both 1K and 15K scales; (2) an MS MARCO-trained cross-encoder reranker consistently reduces precision on scientific text, suggesting domain mismatch outweighs the benefits of stronger query-passage interaction; and (3) RAGAS faithfulness increases with corpus scale. The paper frames itself as a reproducible, controlled comparative study rather than a benchmark and releases code, indexes, and evaluation outputs.

Summary Of Strengths:

1. Clear, focused empirical contribution: The paper asks a well-defined question (how do retrieval and reranking components behave across corpus scales?) and provides a controlled, systematic answer. The scope is appropriately narrow for an empirical study.
2. Compute-aware design: The explicit focus on resource-constrained settings and the three-scale evaluation (1K, 5K, 15K papers) is a practical contribution. The compute budget table (Appendix G) provides useful guidance for practitioners.
3. Negative result is valuable: The finding that an MS MARCO-trained cross-encoder reduces precision on scientific text is important. It highlights the risk of blindly transferring web-trained components to scientific domains and provides empirical evidence for a common intuition.
4. Controlled experimental design: By holding preprocessing, chunking, embedding model, retrieval settings, and generation fixed across scales, the authors isolate the effect of corpus scale. This makes failure modes easier to inspect and results more interpretable.
5. Reproducibility commitments: The authors commit to releasing code, indexes, and evaluation outputs that support replication and follow-up work.
6. Transparent limitations: The paper is honest about its limitations: pseudo-relevance labels, small query set (15 queries), titles/abstracts only, and RAGAS limitations. This transparency is commendable.
7. Practical implications: The findings provide actionable guidance for practitioners building scientific RAG systems: use hybrid retrieval, test rerankers before deployment, and be cautious about web-trained components.

Summary Of Weaknesses:

Pseudo-relevance label circularity: The retrieval evaluation uses pseudo-relevance labels derived from the top-3 hybrid results. This inherently favors the hybrid system and limits the validity of retrieval comparisons. The authors acknowledge this, but it remains a major methodological limitation.

Very small query set: Fifteen evaluation queries are extremely small. This severely limits statistical power and generalizability. The authors acknowledge this, but it undermines confidence in the results.

Limited evidence depth: The paper indexes only titles and abstracts, not full text. This is a significant limitation for scientific QA, where key evidence often appears in methods or results sections. The authors acknowledge this, but it substantially reduces the practical value of the findings.

Limited methodological novelty: The paper does not propose new methods; it evaluates existing components. While empirical studies are valuable, the contribution to *ACL venues is likely to be judged on methodological novelty, which is limited here.

Single dataset focus: The evaluation is on CORD-19 only. Scientific QA encompasses many domains (biomedical, physics, computer science). Generalizability to other scientific corpora is untested.

Single generator: The paper uses only GPT-4o-mini for generation. It is unclear whether the generation findings (faithfulness increasing with scale) hold with other LLMs.

Reranking evaluation limited: The paper reports only precision before/after reranking. It does not report whether reranking changes recall or whether there are cases where reranking helps despite the average decline. A more nuanced analysis would be valuable.

Lack of significance testing: The paper does not report statistical significance for the differences between retrieval systems or for the reranking degradation. This is important given the small query set.

Paper type mismatch: The paper reads as a short report or technical note, but the review is for a potential conference paper. The scope and depth are more appropriate for a workshop or system demonstration paper.

Comments Suggestions And Typos:

Improve retrieval evaluation: The most critical improvement would be to obtain human relevance annotations (even for a subset of queries) to validate the pseudo-label findings. This would substantially strengthen the paper.

Expand query set: Even 50-100 queries would provide more statistical power. The authors should consider augmenting the 15 queries with additional questions or using existing QA datasets (e.g., COVID-QA).

Add significance testing: Report statistical significance (e.g., paired bootstrap or t-tests) for key comparisons. This is essential given the small sample size.

Expand reranking analysis: Report recall@K alongside precision@K for reranking. Explore whether reranking helps for some query types even if it hurts on average.

Consider human evaluation: A small human evaluation of generation quality (beyond RAGAS) would provide additional credibility, given RAGAS limitations.

Analyze failure cases: Include qualitative examples of cases where hybrid retrieval succeeds and where reranking fails. This would make the findings more interpretable.

Explain R@1 denominator: The paper notes that with three pseudo-relevant documents, the maximum R@1 is 1/3 = 0.333. This is an unusual detail; clarify why this metric is still meaningful.

Add more chunking analysis: The chunking analysis in Appendix B is brief. Consider reporting retrieval results with different chunking strategies to show the impact on downstream retrieval.

Typos:

- Page 1: "retrieval- augmented generation" appears twice in the abstract.
- Page 2: "each retriever" should be "each retrieval system" or similar.
- Page 3: The footnote reference "text"[115, 873, 486, 919], [511, 412, 881, 492]] appears to be a placeholder or artifact.
- References: Some citation formatting is inconsistent (e.g., "and 1 others" in some entries).

Clarify CORD-19 version: Specify which version of CORD-19 was used and whether it was filtered (e.g., only peer-reviewed papers).

Mean chunk tokens: Table 1 shows "Mean chunk tokens = 215" across all scales. This is surprising; explain why the mean does not change with scale (same chunking strategy applied to different random samples).

Context precision is low: Table 4 shows context precision is very low (0.095-0.122). Discuss why this is the case and whether it indicates a problem with the retrieval or the RAGAS metric.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 2.5

Excitement: 2.5

Overall Assessment: 2.5 = Borderline Findings

Best Paper Justification:

N/A — My overall assessment is "Borderline Findings" (2.5), not "Consider for Award" or "Borderline Award". Therefore, this section is not applicable.

Limitations And Societal Impact:

Strengths:

- The paper includes a dedicated Limitations section that honestly discusses: pseudo-relevance labels, small query set, titles/abstracts only, RAGAS limitations, and lack of human evaluation.
- The ethics statement notes that generated answers should not be used for medical decision-making without expert review.
- The generative AI disclosure is transparent about using Claude and ChatGPT for prose refinement.

Missing or insufficient points:

- Generalizability: The paper does not discuss whether findings generalize beyond CORD-19 or beyond COVID-19 literature. Scientific QA in other domains (physics, chemistry, computer science) may behave differently.
- Failure mode analysis: The paper does not analyze specific failure cases or identify systematic biases in the retrieval or generation.
- Clinical implications: While the paper warns against medical decision-making, it does not discuss the broader societal implications of automated scientific QA systems (e.g., reinforcing existing biases in the literature, over-reliance on automation, misinformation risks).
- Data bias: CORD-19 papers are predominantly English-language and from specific journals. The system may not generalize to non-English scientific literature or to literature from underrepresented regions.
- Equity considerations: The paper does not discuss who benefits from scientific RAG systems or who might be excluded (e.g., researchers with limited access to compute, practitioners in low-resource settings).

Societal impact considerations: The system is intended to assist with scientific question answering but has clear limitations (titles/abstracts only, pseudo-labels). The authors appropriately frame it as a research prototype rather than a deployable system. However, the paper should more explicitly discuss how the limitations could lead to harmful outcomes if the system is used without expert oversight.

Ethical Concerns:

None. The paper uses public data (CORD-19) and does not involve human subjects. The authors are transparent about the limitations and include a generative AI disclosure. The paper does not appear to violate the ACL Code of Ethics.

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 4 = Useful: I would recommend the new datasets to other researchers or developers for their ongoing work.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A. I do not know anything about the paper from outside sources.

Knowledge Of Paper Source: N/A. I do not know anything about the paper from outside sources

Knowledge Of Paper Source Other: N/A

Impact Of Knowledge Of Paper: N/A. I do not know anything about the paper from outside sources

Knowledge Of Paper Additional: N/A

Knowledge Of Authors Guess: N/A

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I did not use any generative AI tools for this review



Official Review of Submission8722 by Reviewer VbPC

Official Review by Reviewer VbPC 30 Jun 2026, 01:21 (modified: 08 Jul 2026, 18:56) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer VbPC, Authors

Revisions

Paper Summary:

This paper examines a controlled evaluation of retrieval strategies for scientific retrieval-augmented generation (RAG) using the CORD-19 corpus. It compares sparse, dense, and hybrid retrieval methods, examines the performance of a general-domain reranker in the scientific domain, and evaluates generation quality using RAGAS. Experimental results show that hybrid retrieval consistently outperforms individual retrieval methods, while the general-domain reranker degrades retrieval performance and larger literature collections improve response quality.

Summary Of Strengths:

By keeping the retrieval and generation pipeline fixed while varying only the literature collection size, the study clearly isolates the impact of corpus size and enables a fair comparison of retrieval

Summary Of Weaknesses:

1. Using only 15 queries for retrieval and generation evaluation makes it difficult to draw statistically reliable conclusions or assess the significance of the reported improvements.
2. Only a single general-domain MS MARCO cross-encoder is evaluated, making it difficult to determine whether the observed performance degradation is caused by domain mismatch or by the specific reranker itself.
3. Although the paper reports several RAGAS metrics, it does not adequately explain their definitions or discuss their limitations. Moreover, relying solely on automated LLM-based evaluation without human annotation or fact verification reduces the credibility of the reported generation quality.

Comments Suggestions And Types:

1. Expand the evaluation to a substantially larger query set and report statistical significance tests where appropriate. Incorporating manually annotated relevance labels would further improve the reliability of the retrieval evaluation.
2. Include additional reranker baselines, particularly domain-adapted or scientific rerankers, to better distinguish between the limitations of general-domain rerankers and the effects of cross-domain transfer.
3. Provide a clearer description of the RAGAS metrics and complement automated evaluation with human annotation or factual verification. Combining automatic and manual evaluation would improve the reliability of the reported generation quality.

Confidence: 5 = Positive that my evaluation is correct. I read the paper very carefully and am familiar with related work.

Soundness: 2 = Poor: Some of the main claims are not sufficiently supported. There are major technical/methodological problems.

Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the ACL community.

Overall Assessment: 2 = Resubmit next cycle. I think this paper needs substantial revisions that can be completed by the next ARR cycle.

Ethical Concerns:

There are no concerns with this submission

Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A. I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A. I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A. I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

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